The documentation and process conversion measures necessary to comply with this document shall be completed by 5 June 2020.

INCH-POUND

MIL-PRF-19500/296G
6 February 2020
SUPERSEDING
MIL-PRF-19500/296F
20 January 2012

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD-EFFECT, P-CHANNEL, SILICON, THROUGH-HOLE AND SURFACE MOUNT PACKAGES, TYPE 2N2609, QUALITY LEVEL JAN

Inactive for new design for the 2N2609 device after 7 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for P-channel, junction, silicon field-effect transistors. One level of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Package outlines. The device package outlines are a modified DO-206AA (formerly a modified TO-18) in accordance with figure 1 and an unleaded surface mount (UB) in accordance with figure 2.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25 °C.

Types	P_T (1) T_A = +25 °C	V_{GSS}	T_{STG} and T_J
	mW	V dc	°C
2N2609, 2N2609UB	300	30	−65 to +200

(1) Derate linearly, 1.71 mW/°C for T_A = +25 °C.

1.4 Primary electrical characteristics. Unless otherwise specified, at T_A = +25 °C.

Limit	I_{DSS} V_{DS} = −5 V dc V_{GS} = 0	$V_{GS(off)}$ V_{DS} = −5 V dc I_D = −1.0 µA dc	C_{lss} V_{DS} = −5 V dc V_{GS} = 0 V dc f = 1 MHz	$\lvert Y_{FS} \rvert$ V_{DS} = −5 V dc V_{GS} = 0 V dc f = 1 kHz	I_{GSS} V_{GS} = 15 V dc V_{DS} = 0 V dc
	mA dc	V dc	pF	µmho	nA dc
Minimum	−2.0	0.75		2,000	
Maximum	−10.0	6.00	10	6,250	22.5

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AMSC N/A

FSC 5961

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. Quality level designator "JAN" is the only level that is applicable for this specification sheet.

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The only second number symbols for the device covered by this specification sheet are "2609".

1.5.3 Suffix symbol. Devices which use no suffix symbol indicate a DO-206AA (formerly a modified TO-18) (see figure 1). The suffix letters "UB" are used on devices to indicate a 4 pad surface mount package. The metal lid is connected to pad 4 (see figure 2).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

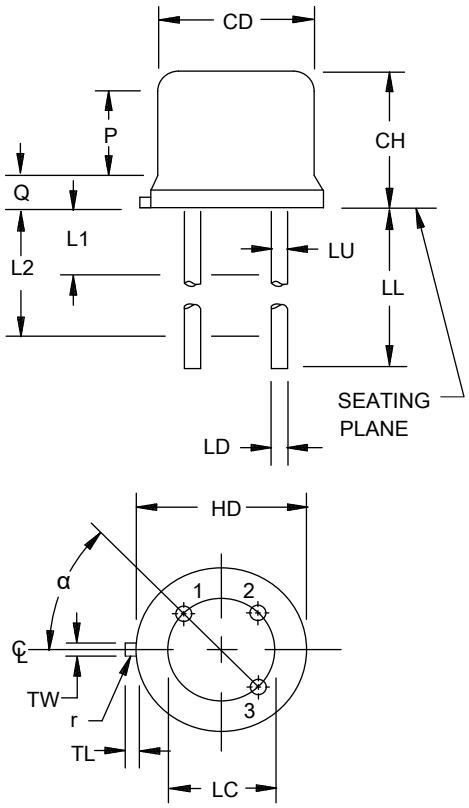
MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	3
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	3
LC	.100 TP		2.54 TP		4
LD	.016	.021	0.41	0.53	5, 6
LL	.500	.750	12.70	19.05	5, 6
LU	.016	.019	0.41	0.48	5, 6
L1		.050		1.27	5, 6
L2	.250		6.35		5, 6
P	.100		2.54		
Q		.030		0.76	3
TL	.028	.048	0.71	1.22	7
TW	.036	.046	0.91	1.17	8
r		.010		0.25	9
α	45° TP		45° TP		4

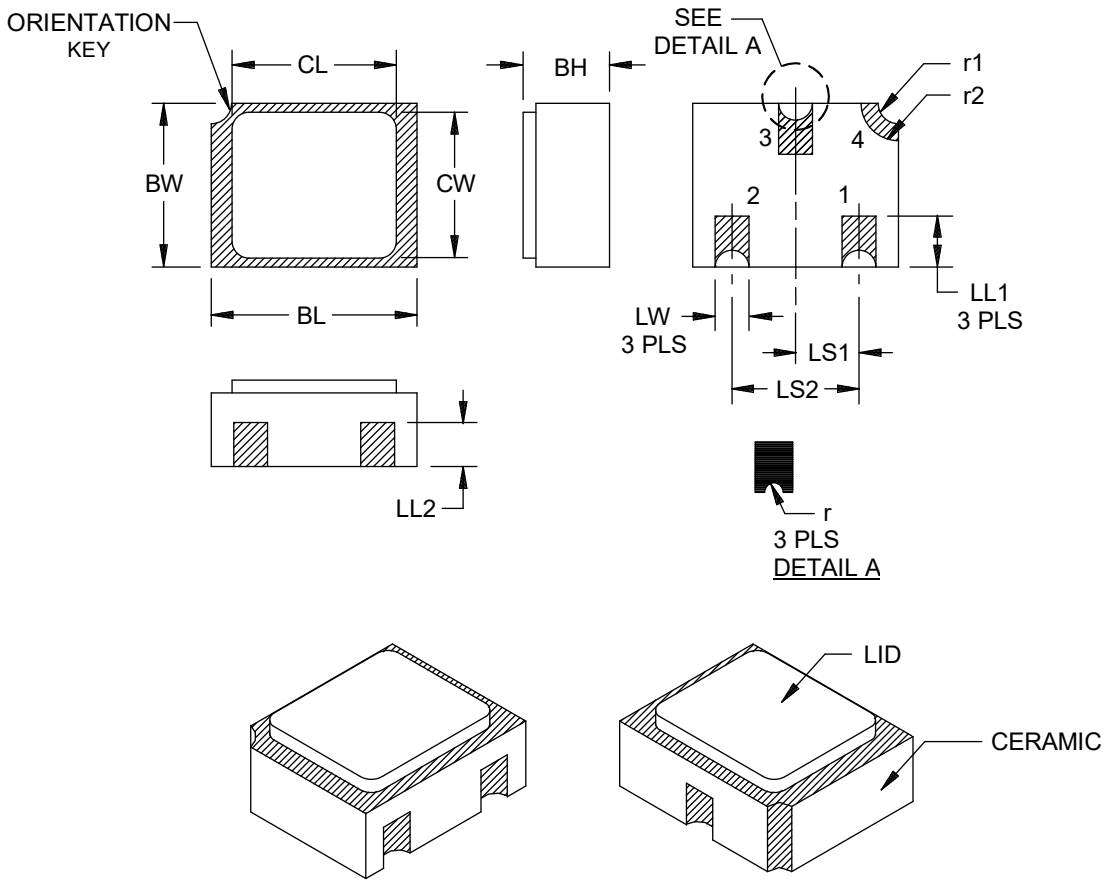


NOTES:

1. Dimension are in inches. Millimeters are given for general information only.
2. Terminal 1 = source, terminal 2 = gate, terminal 3 = drain. The gate shall be electrically connected to the case.
3. Body contour optional within zone defined by dimensions CD, HD, and Q.
4. Leads at gauge plane .054 +.001 −.000 inch (1.37 +0.03 −0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
5. Dimension LU applies between dimensions L1 and L2. Dimension LD applies between dimensions L2 and LL minimum. Diameter is uncontrolled in dimension L1 and beyond dimension LL minimum.
6. All three leads (see 3.4.1).
7. Dimension TL measured from maximum HD.
8. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 inch (0.28 mm).
9. Dimension r (radius) applies to both inside corners of tab.
10. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for modified TO-206AA package (2N2608).

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Symbol	Dimensions				Note	Symbol	Dimensions				Note
	Inches		Millimeters				Inches		Millimeters		
	Min	Max	Min	Max			Min	Max	Min	Max	
BH	.046	.056	1.17	1.42		LS1	.035	.040	0.89	1.02	
BL	.115	.128	2.92	3.25		LS2	.071	.079	1.81	2.01	
BW	.085	.108	2.16	2.74		LW	.016	.024	0.41	0.61	
CL		.128		3.25		r		.008		0.20	
CW		.108		2.74		r1		.012		0.31	
LL1	.022	.038	0.56	0.96		r2		.022		0.56	
LL2	.014	.035	0.36	0.89							

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Hatched areas on package denote metallized areas.
3. The lid material shall be Kovar.
4. Pad 1 = Drain, pad 2 = Source, pad 3 = Gate, pad 4 = Shielding connected to the lid.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for UB surface mount package (2N2609UB).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Pin-out. The pin-out of the devices shall be as shown on figures 1 and 2 as applicable.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated PIN, the date code, and the manufacturers symbol or logo. The JAN prefix can be abbreviated as "J". The "2N" prefix and the "UB" suffix can also be omitted.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein shall be performed by the first inspection lot of this revision to maintain qualification.

4.3 Screening. Screening is not applicable for devices of quality level JAN for this specification sheet.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

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4.3.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB JAN of MIL-PRF-19500 and herein. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
B3	1027	$V_{CB} \geq 10$ V dc, $T_A = +150$ °C ±5 °C, $V_{DS} = 0$; $V_{GS} = 24$ V dc. No heat sink or forced-air cooling on the devices shall be permitted.
B3	2037	Bond strength, condition D.
B5		Not applicable.
B6	1032	$T_{STG} = +200$ °C.

4.3.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	2036	Test condition A.
C6	1026	Steady-state operation life (accelerated): 1,000 hours at $V_{GS} = 24$ V dc, $T_A = +150$ °C ±5 °C, $V_{DS} = 0$. No heat sink or forced-air cooling on the devices shall be permitted.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with the applicable steps of table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Gate - source breakdown voltage	3401	Bias condition C; I_G = 1.0 µA dc; V_{DS} = 0	$V_{(BR)GSS}$	30		V dc
Gate reverse current	3411	Bias condition C; V_{GS} = 30 V dc; V_{DS} = 0	I_{GSS1}		30	nA dc
Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS2}		22.5	nA dc
Subgroup 2 - continued						
Zero-gate-voltage drain current	3413	Bias condition C; V_{DS} = 5 V dc; V_{GS} = 0, pulsed (see 4.5.1).	I_{DSS1}	−2.0	−10.0	mA dc
Gate to source cutoff voltage	3403	I_D = 1 µA dc; V_{DS} = 5 V dc	$V_{GS(off)}$	0.75	6.0	V dc
Subgroup 3						
High-temperature operation:		T_A = +150 °C				
Drain current	3413	Bias condition C, V_{GS} = 0 V dc; V_{DS} = 5	I_{DDSS2}	−1.0		mA dc
Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0.	I_{GSS3}		45	µA dc
Low-temperature operation:		T_A = −55 °C				
Small-signal, common-source, short-circuit, forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 kHz.	$\lvert Y_{fs1} \rvert$		9,375	µmho
Subgroup 4						
Small-signal common-source, short-circuit, forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 kHz.	$\lvert Y_{fs2} \rvert$	2,000	6,250	µmho

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 − continue						
Small-signal common-source, short-circuit, input capacitance	3431	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 MHz.	C_{iss}		10	pF
Common-source spot noise figure		V_{DS} = 5 V dc; V_{GS} = 0; B_W = 16 percent; R_G = 1 Mohm; e_{gen} = 1.82 mV; R_L = 220 ohms; f = 1 kHz; (see figure 3)	NF		3	dB
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups B, C, and E electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1	Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS1}		23	nA dc
2	Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS2}		45	nA dc
3	Drain current	3413	Bias condition C; V_{GS} = 5 V dc; V_{DS} = 0	I_{DSS1}	−2.0	−10.0	mA dc
4	Drain current	3413	Bias condition C; V_{GS} = 5 V dc; V_{DS} = 0	I_{DSS2}	−1.80	−11.0	mA dc
5	Small-signal, common-source short-circuit, forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 kHz	$\lvert Y_{fs1} \rvert$	2,000	6,250	µmho
6	Small-signal, common-source short-circuit, forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 kHz	$\lvert Y_{fs3} \rvert$	1,800	6,875	µmho

1/ The electrical measurements for table E-VIB of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, steps 1, 3, and 5.
   b. Subgroups 3 and 6, see table II herein, steps 2, 4, and 6.

2/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroups 2 and 3, see table II herein, steps 1, 3, and 5.
   b. Subgroup 6, see table II herein, steps 2, 4, and 6.

3/ The electrical measurements for table E-IX of MIL-PRF-19500 are: Subgroups 1 and 2, see table II herein, all steps.

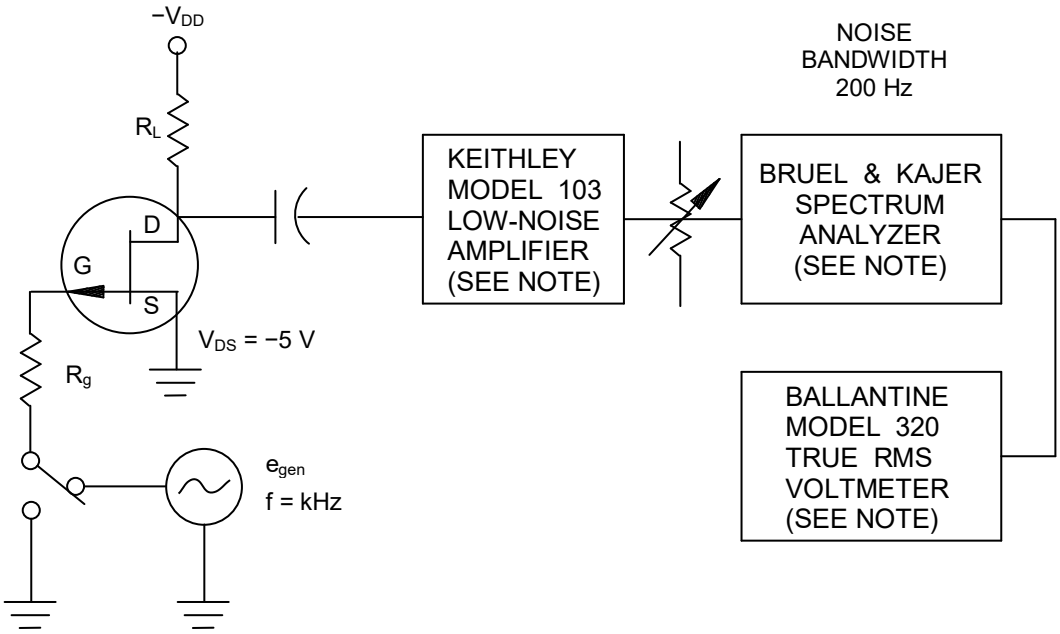
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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling	1051	−55 °C to +150 °C, 500 cycles	
Hermetic seal	1071	Fine and gross leak	
Electrical measurements		See table II, all steps	
Subgroup 2 1/			45 devices, c = 0
Steady-state operating life	1026	V_{CB} ≥ 10 V dc, T_A = +150 °C, V_{DS} = 0; V_{GS} = 24 V dc.	
Electrical measurements		See table II, all steps	
Subgroups 4 and 5			
Not applicable			
Subgroup 8			45 devices, c = 0
Reverse stability	1033	Condition A	

1/ A separate sample for each test shall be pulled.

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NOTE: Or equivalents

Procedure: The voltage and current shall be applied to the terminals, and the noise figure shall then be measured as follows:

1. Connect e_{gen} to the input of the device under test.
2. Adjust gain of system to give 0 dB reading on a convenient scale of the Ballintine voltmeter or equivalent.
3. Switch the device input to ground.
4. Increase system gain by 60 dB.
5. Read noise figure directly in dB's on Ballintine scale or equivalent.

FIGURE 3. Common-source spot noise figure test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

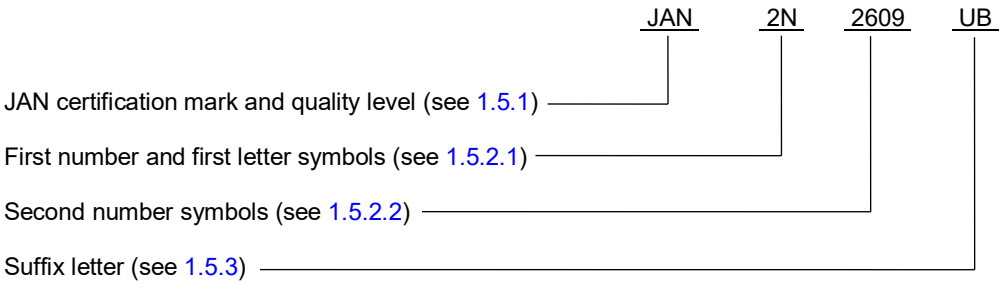
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices and are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available for devices covered by this specification sheet.

Quality level "JAN"	JAN2N2609	JAN2N2609UB	

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6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army − CR
Navy − EC
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AV, MI, SM
Navy − AS, MC, OS, SH
Air Force − 19

Preparing activity:
DLA − CC

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